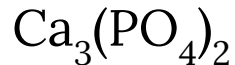
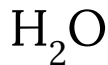


Empirical and Molecular Formulas

Share one copy of this with your group.
Make your own copy at the end of the lesson.

Recall: Breaking Down Compounds into Elements

Each person in the group should work on one of the following compounds:



Guiding Questions:

1. How many atoms of each element are in your compound?
2. What is the molar mass of each element in your compound? (Use the periodic table.)
3. How do you calculate the total molar mass of the compound?

If unsure of your answers, check with another group &/or instructor.

ESSENTIAL QUESTIONS

- How can we determine what fraction of a compound's mass comes from each of its elements?
- Why is knowing the percentage composition of a compound useful in real-world applications?
- How does percentage composition help us predict the simplest (empirical) formula of a compound?

Remember, you should be able to thoroughly answer these AFTER learning through following resources.

Instructions

- Follow through the Engage to Apply resources in order .
- Once confident, attempt Taking it Further work if available. This is Level 4 for course.
- Record your learning in these slides. Remember to keep track of where to find these slides for studying!
- Submit work from the Apply Section, IF INDICATED. Otherwise, it is regular practice work.
- Try the Extend section as desired to deepen your understanding

Links To sections

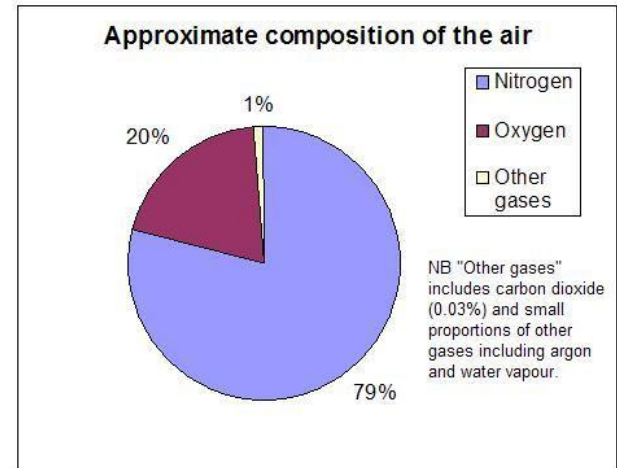
Click for quick access to:

- [Engage](#)
- [Explore](#)
- [Apply](#)
- [Extend](#)

Part 1: ENGAGE (5 minutes)

Engage: Using percents!

- In your group, answer the following:
- What does percent mean?
- Using the pie graph, if you had 100L of air, how much would be nitrogen?
- If you had 50L of air, how much would be oxygen?
- If you had 34.6L of air, what equation could you set up to solve for the amount of nitrogen?



Engage: Scenarios

Discuss how knowing the percentage composition of a substance would be important in at least 2 of the following:

- Pharmaceuticals/Medicines
- Crime Scene situations
- Chemical spills
- Agricultural fertilizers
- Water pollution science

Part 2: EXPLORE (20 - 30 minutes)

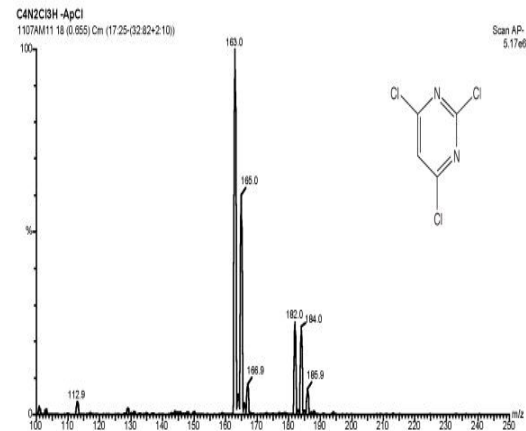
Explore: How we determine composition

RECALL: Law of Constant Composition means compounds contain elements in fixed proportions that are Always the SAME for the same compound!

Equipment for determining the formula (proportions):

1. Mass Spectrometer (as seen on CSI shows): can determine the molar mass of many compounds by vapourizing a small sample, ionizing it (giving it a charge) and deflecting the charged gas in a magnetic field. The amount of deflection gives the mass.
2. Combustion Analyzer: Determines amount of C, H, O and N by burning the substance in pure oxygen and analyzing the products.

**** Be able to recognize these two methods for multiple choice questions.**



Explore: Definitions to Know

Empirical Formula

- Gives the lowest ratio of atoms in a compound, but may not tell the exact number of each type of atom

Molecular Formula

- Indicates the actual numbers of atoms in one molecule of a compound

Explore: What is the matching empirical formula?

	Molecular	Empirical
Hydrogen peroxide	H_2O_2	Fill in!
Sugar	$\text{C}_6\text{H}_{12}\text{O}_6$	Fill in!

Check your work on the next slide.

Explore: Check your understanding!

	Molecular	Empirical
Hydrogen peroxide	H_2O_2	HO
Sugar	$\text{C}_6\text{H}_{12}\text{O}_6$	CH_2O

Explore: A good problem solving strategy!

If you aren't already, please USE GRASP in Chemistry:

Given: List all information with units given in problem.

Required: Include what you need to determine in your data list

Analysis: List formulas first, then enter given data

Solution: solve for Required. There may be more than one. Remember UNITS.

Paraphrase: A therefore statement indicating your understanding of the solution.

Explore: Percentage Composition

The next few slides will ask you to think through calculating the percentage composition of compounds. First, from mass data such as that generated by mass spectrometers. Then from atomic mass of the elements in a compounds.

REMEMBER: Percent is just percent! How much of one element out of the total mass of the compound?

Show your work!

Explore: Try this!

Complete next several slides on a class whiteboard and insert photos of your work (or other record) on each slide.

Solve by working through next several slides: A 13.5g sample of a compound is determined to have 7.46g of strontium and 6.04 of chlorine. Determine the % composition of each element.

Explore: Problem 1 - Calculate the % composition of carbon dioxide, CO_2 from atomic mass.

Step 1 : Calculate total atomic mass of each element in the compound.

$$m_c =$$

$$m_o =$$

Explore: Problem 1

Show your calculation of the molecular mass of CO₂.

Molecular mass_{CO₂} =

Ans: 44.01amu (or just
u for units)

Explore: Problem 1 - Now determine % composition by mass of oxygen in CO₂. Then do % of carbon. {Use GRASP}

Therefore statement:

Explore: Problem 2 - Empirical formulas from Mass Data

A compound contains the following % compositions:

$$\text{C} = 38.71\%$$

$$\text{H} = 9.71 \%$$

$$\text{O} = 51.58\%$$

Determine the mole ratio of the elements. Use the mole ratio to determine the Empirical Formula.

HINT: If you have a 100g sample, how many grams will be carbon? If you know the grams, what quantity can you convert to? Check the next blue solution slide to get started after trying!

Solution:

100g sample

$$\text{C} = 38.71\% \rightarrow 38.71\text{g}$$

$$\text{H} = 9.71\% \rightarrow 9.71\text{g}$$

$$\text{O} = 51.58\% \rightarrow 51.58\text{g}$$

Convert each to moles

$$n = m/\text{MM}$$

$$\begin{aligned} n_{\text{C}} &= 38.71\text{g}/12\text{g}\text{mol}^{-1} \\ &= 3.22\text{ mol} \end{aligned}$$

$$\begin{aligned} n_{\text{H}} &= 9.71\text{g}/1.01\text{g}\text{mol}^{-1} \\ &= 9.61 \end{aligned}$$

$$\begin{aligned} n_{\text{O}} &= 51.58\text{g}/16.00\text{g}\text{mol}^{-1} \\ &= 3.22 \end{aligned}$$

You aren't at a chemical formula yet, but you do have mole ratios.

Discuss as a group how you could use them to determine the empirical formula (lowest whole number ratio), then continue to the next slides for some guidance!

Explore: Problem 2 - Empirical formulas from Mass Data

You now have a molar ratio of the elements:

C : H : O

3.22: 9.61 : 3.22

Numbers in compound formulas are always whole numbers though. What can you do to get whole numbers? Think before going to next slide!

Solution

C: H : O

3.22: 9.61 : 3.22

Divide by smallest # to get to whole number ratio. Round the 2.98 to 3. Recall this was experimental data so it won't give perfect results. 2.98 is very close to 3 within experimental error!

Final empirical formula: CH_3O

Explore: Problem 3 - Determine molecular formula given the molecular mass of the compound.

Reminder: Empirical formula is CH_3O . Recall the difference between empirical and molecular formulas!

Solve: If the molecular mass is 93.12 g/mol, what is the molecular formula?

Discuss how you could use this information together with the empirical formula and try to solve. Check in with your instructor to see if you are correct!



Explore: Problem 4 - Determine the formula, using the same process, for a compound containing 69.9% Fe and 30.1% O.

Explore: Problem 4 – If you found a ratio of 1: 1.5 for Fe:O (this is a correct step towards the final answer):

Since 1.5 atoms of oxygen isn't possible, how can this ratio be maintained in the final empirical formula without having 0.5 of an atom? Show your work!



Part 3: Apply
(20-40 minutes; includes home practice time)

Apply: Check your understanding

1. A compound is 27.5% carbon and 72.5.0% oxygen by mass. What is its empirical formula?

- a) CO
- b) CO₂
- c) C₂O
- d) C₃O₂

Room to show your work:

Answer: move box to check

Apply: Check your understanding

2. If 100g of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) is analyzed, how many grams of hydrogen are present?

- a) 1.2g
- b) 22.0g
- c) 13.0g
- d) 6.5g

Room to show your work:

Answer: move box to check

Apply: Check your understanding

3. A student gets an empirical formula of $\text{CH}_{1.32}$. How can they fix this?

Group answer:

4. Why do $\text{C}_6\text{H}_{12}\text{O}_6$ (glucose) and $\text{C}_3\text{H}_4\text{O}$ (cyclopropanone) behave differently despite having the same elements?

Group answer:

Answer: move box to check

Apply: Real world

Why is percentage composition used in forensic chemistry?

- a) To measure a compound's density
- b) To identify substances by comparing elemental ratios
- c) To determine the color of a powder
- d) To calculate the boiling point of liquids

Answer:

Apply: Practice Work

Complete the practice work as indicated in the unit plan and/or sheets given in class.

If time allows you can ask for and play “Kitchen or Crime Scene”

Remember! Its important in this unit to do plenty of practice to be confident of each topic before moving to the next one! It is also good practice to save some practice work, once you are confident, to complete over the next several days & weeks from the lesson. Spaced practice solidifies your understanding and long term retention!

Part 5: Beyond the course extensions & Extra help {optional}

Extend: Read this article. On the spot crime scene analysis!

[MIT develops handheld mass spectrometer.](#)